

RGB, 3 views

"Put the pear in the basket."

depth + robot state

Frozen VLA

$\pi_{0.5}$, no update

noun attn.

features "pear"

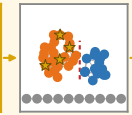
image action

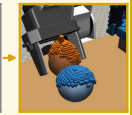
KV

ATS which object, where it ends



seed patches $(\Lambda + L)m = \Lambda y$





surface $\mathcal{T}_t$

same object?



raw centered

target

other

low → alert

HSR when its role changes

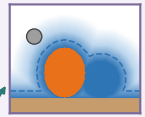
phase: pick → place → home

grasp: free → held → free

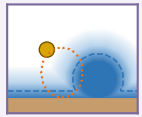
probe + gripper + geometry

LTE who may touch it

cuRobo TSDF → ESDF



d_t other links



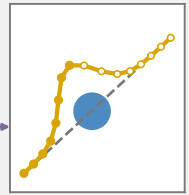
$d_t^{\mathcal{X}}$ contact links

$\delta_i = d_t^{\mathcal{X}}$ iff $i \in \mathcal{L} \wedge e_t$



after grasp: object becomes attached spheres

CCR



refine A_t

s.t. $h_{ik} \geq 0$

run first K

check prefix

execute hold

action chunk A_t (policy output, unchanged)